Louis Pasteur and Sir Frank Macfarlane Burnet

Louis Pasteur

Louis Pasteur (1822–1895) was a pioneer in the search for the causes of disease in the second half of the 19th century. His work included:

- how to kill bacteria in wine and beer without spoiling the flavour of the beverage
- disproving theory of 'spontaneous generation'
- discovery of bacilli causing disease in silkworms
- discovery of inoculation against the bacteria that cause anthrax.

1 Define *beverage*.

Louis Pasteur was the founder of the science we now call *microbiology*. Microbiology is the study of microscopic forms of life. In 1854 Pasteur moved to Lille, France, to take up a post at the local university. One of his projects was to investigate why some beer and wine went sour – a problem that cost the French manufacturers of these products a great deal of money. The manufacture of both beer and wine involves the changing of carbohydrates into ethanol (alcohol) in a process called fermentation.

Pasteur examined many fermenting mixtures under the microscope and found different kinds of microbes. He demonstrated that yeast was responsible for forming ethanol and that other microorganisms (bacteria) caused the production of ethanoic (acetic) acid – the substance causing the sour taste. Vinegar, which is mainly ethanoic acid, is named from the French *'vinaigre'* meaning 'sour wine'.

2 The bacteria causing ethanol to change to ethanoic acid are *Acetobacter*, a type of anaerobic bacteria. What is meant when bacteria are anaerobic?

3 What is the chemical name for vinegar?

Pasteur stated that fermentation and decay were both caused by microorganisms and if these microorganisms could be killed after fermentation had occurred in wine, the wine would not go sour. His first attempts to do this involved boiling the wine, which, although it killed the microbes, also spoiled the flavour. After many experiments, he found that a good compromise was to heat the wine to around 60°C for a short period of time. This process (pasteurisation) is still used to kill bacteria in milk and beer but small amounts of sulfur dioxide are now added to wine to act as a preservative instead of heating it.

4 Does pasteurisation kill all of the microbes in milk? Explain your answer.

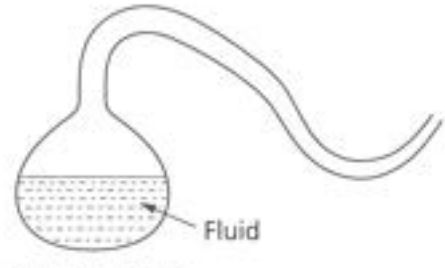
Some scientists in Pasteur's time carried out experiments that involved boiling hay with water under sterile conditions and these, if left for a few days, generated large numbers of microorganisms. They used these results to support the theory of spontaneous generation of life. Louis Pasteur carried out similar experiments using sugar and yeast mixtures boiled in water. He found that only those mixtures exposed to the air grew microorganisms.

5 Explain what is meant by *sterile conditions*.

6 Compare the difference between an *aerobic* organism and an *anaerobic* organism.

7 What sort of organism was found in the boiled hay water – aerobic or anaerobic? Explain how you came to this answer.

In 1864 Pasteur gave a demonstration to the French Academy of Science. He prepared 60 glass flasks containing infusions (mixtures) of water, yeast and sugar. The contents were boiled for two minutes to kill anything present and all but four of the flasks were sealed. These four flasks had the necks melted by heat and drawn out into a long curve, with the necks remaining open to the atmosphere. Pasteur then boiled the contents and let steam come out of the neck of each flask in order to kill any germs present.



Gooseneck flask

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The remaining flasks were then split into three groups. The first group was opened briefly and then resealed near where the scientists were holding their meeting. The second group was opened and resealed high in the same building. The third group was opened and resealed outside the building. The flasks were left for a few days.

The four flasks with open, curved glass necks (gooseneck flask) remained sterile. All other groups had some flasks with microorganisms growing and some remaining sterile. Pasteur's hypothesis was that microbes from the air needed to reach the contents of the flask before microorganisms could grow, and microbes from the air entering the curved necks of the open flasks could not travel through to the nutrients.

8 Why were the contents of the flask boiled?

9 Explain the difference between fermentation and decay.

10 What is an infusion? Give an example.

11 Suggest why the groups of flasks described above were exposed to air under different conditions.

Anthrax and Pasteur

Anthrax is a contagious disease of cattle and sheep (and other warm-blooded animals, including humans). In Pasteur's time, it caused the deaths of large numbers of livestock and occasionally humans. The cause of the disease is *Bacillus anthracis.* Pasteur was not the first person to develop a culture for this type of bacteria (Robert Koch was the first, in 1863), but he was the first person to develop a vaccine for anthrax.

Pasteur developed a method of inoculating with attenuated (weakened) cultures of germs to give immunity against disease. He publicly demonstrated, with a flock of sheep, that vaccinated animals survived a subsequent infection of anthrax and unvaccinated animals had an almost 100% mortality rate when infected with anthrax.

Pasteur had to work hard to overcome the prejudices of many doctors and scientists of his time, since it meant proving what many of them believed to be wrong. They could not believe that tiny organisms such as bacteria could kill people, but thought that the role that germs played (if they existed at all) was not of any great importance. Pasteur played a major part in the gradual acceptance by the medical community of the fact that diseases could be caused by bacteria.

12 Define *culture* in scientific terms.

13 Define *vaccine.*

14 What is meant by 'an almost 100% mortality rate'?

15 What is meant by 'immunity against disease'?

Pasteur was one of the main proponents of the 'germ theory of disease', the theory that diseases are caused by the body being attacked by microorganisms. People found it hard to believe that organisms, many of which could not be seen by the naked eye, could kill people. This was also contradictory to the generally held belief of the time in *'spontaneous generation'* of life. At the time, many people thought that simpler forms of life, such as insects, could arise from inorganic substances under certain circumstances – *'abiogenesis'.*

16 Which invention helped to disprove *abiogenesis*?

17 What does this define: *'The use of heat to kill pathogens in a liquid without causing major changes to its properties, eg taste and appearance'*?

18 Name two liquids that are pasteurised.

19 Which organism is used to ferment sugars to make beer and wines?

20 Name another food that is prepared using fermentation.

21 Which disease is caused by *Bacillus anthracis*?

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22 One of Pasteur's favourite quotations was: *'Where observation is concerned, chance favours only the prepared mind.'* Rewrite Pasteur's quotation in your own words.

23 Explain how a gooseneck flask can stop microorganisms from the air coming into contact with the contents.

24 Vaccines can be formed using dead strains of an infective agent then injecting the dead organism into the body. How might this be of use in helping to protect an animal from the live, infective form?

25 With reference to question 24, why might it be impossible to inject small amounts of certain *live* infective agents into a body instead? Include an example of an infective agent that could not be injected in its live form.

Research

26 Who was the first person to see microorganisms?

27 Outline Pasteur's work in the discovery of bacteria causing disease in silkworms.

28 Pasteur was not the first person to inoculate against disease. This was thought to have been William Jenner in around 1798. Give an outline of Jenner's work in vaccination.

Sir Frank Macfarlane Burnet

Sir Frank Macfarlane Burnet (1899–1985) was an Australian (born in Traralgon, Victoria) *virologist.* A graduate from the University of Melbourne, he worked as a doctor, *pathologist* and ship's surgeon. He gained a doctorate (PhD) in London and, from 1928, became the Assistant Director of the Institute of Medical Research. By 1944 Sir Frank was the Director of Experimental Medicine at University of Melbourne (1946–1965).

He made a great contribution to our knowledge of viruses. He (with Peter Medawar) was awarded the 1960 Nobel Prize for *Physiology* for discovery of acquired *immunological* tolerance to tissue transplants. Burnet's predictions concerning the ability of the body to accept transplanted tissue were confirmed in *graftings* conducted by Medawar on embryo rats inoculated with foreign tissue.

When he returned from the UK in 1928, Burnet studied *bacteriophages* and later turned his attention to viruses. He did important work on the use of chicken *embryos* for the isolation and analysis of animal viruses. These embryos were used to develop *vaccines.* His major work was the *clonal* selection of antibody production. This theory helped to explain how the body could tell the difference between entities as either 'self' or 'non-self'. Sir Frank discovered that antibodies were not *inherited,* but were produced during foetal development.

29 Write the meanings of the words in *italics* in the passage above.

30 Describe what *acquired immunological tolerance* means.

31 Suggest how the chicken embryos mentioned above might be used to develop a vaccine.

32 Why might a scientist use embryos instead of fully grown specimens?

33 Give an example of a 'self' cell and a 'non-self' cell.

34 Briefly describe how a bacteriophage differs from a virus.

35 Why is it important for tissue transplants to be tolerated by the immune system?

36 We all know that humans are animals yet people often draw an invisible dividing line between 'human animals' and other animals. Nonetheless, humans are often infected by diseases of other animals. Give an example of a disease humans can catch from other animals. Include its symptoms, how it is contracted and possible treatment.

Research

37 Give the name and details of another scientist who has won a Nobel Prize for Physiology.

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